

Standalone Mode.pdf
PDF

provide each calculation clearly

Below are the key calculations from your uploaded paper explained step-by-step clearly (no steps skipped). I've grouped them by section for clarity.

Source

From your uploaded paper:

"A Single Input Variable FLC for DFIG-Based WPGS in Standalone Mode"

 Standalone Mode

1. Wind Turbine Power Calculation

Formula (Eq. 1)

$$P_m = 0.5 C_p \rho \pi r^2 V_w^3$$

Given:

- $C_p = 0.48$
- $\rho = 1.1514 \text{ kg/m}^3$
- $V_w = 12 \text{ m/s}$
- $P_m = 4.81 \text{ kW} = 4810 \text{ W}$

Step-by-step:

$$4810 = 0.5 \times 0.48 \times 1.1514 \times \pi r^2 \times (12)^3$$

$$4810 = 0.5 \times 0.48 \times 1.1514 \times \pi r^2 \times 1728$$

$$4810 = (0.24 \times 1.1514 \times 1728) \times \pi r^2$$

$$4810 = (477.3) \times \pi r^2$$

$$r^2 = \frac{4810}{477.3\pi}$$

$$r^2 \approx \frac{4810}{1499} \approx 3.21$$

$$r \approx \sqrt{3.21} \approx 1.8 \text{ m}$$

✓ Result:

$$r \approx 1.8 \text{ m}$$

✓ 2. Tip Speed Ratio Relation

Formula (Eq. 3)

$$\lambda = \frac{\omega_T r}{V_w}$$

Rearranged:

$$\omega_T = \frac{\lambda V_w}{r}$$

✓ 3. Gear Ratio Calculation

Formula (Eq. 4)

$$\omega_r = \eta \omega_T$$

Substitute ω_T :

$$\omega_r = \eta \cdot \frac{\lambda V_w}{r}$$

Given:

- $\omega_r = 204 \text{ rad/s}$
 - $\lambda = 8.1$
 - $V_w = 12$
 - $r = 1.8$
-

Step-by-step:

$$204 = \eta \cdot \frac{8.1 \times 12}{1.8}$$

$$204 = \eta \cdot 54$$

$$\eta = \frac{204}{54} \approx 3.77$$

✓ Gear ratio:

$$\eta \approx 3.77$$

✓ 4. Reference Rotor Speed (MPPT)

Formula (Eq. 7)

$$\omega_r^* = \frac{\eta \lambda_{opt} V_w}{r}$$

Substitute values:

$$\omega_r^* = \frac{3.77 \times 8.1 \times V_w}{1.8}$$

$$\omega_r^* = \frac{30.537 \times V_w}{1.8}$$

$$\omega_r^* \approx 16.96 V_w \approx 17 V_w$$

✓ Final:

$$\omega_r^* \approx 17 V_w$$

✓ 5. Rated Stator Power of DFIG

Formula (Eq. 9)

$$P_{rs} = \frac{P_{in}}{1 + |s|}$$

Given:

- $P_{in} = 4.81 \text{ kW}$
 - $s = -0.3 \Rightarrow |s| = 0.3$
-

Step-by-step:

$$P_{rs} = \frac{4.81}{1 + 0.3}$$

$$P_{rs} = \frac{4.81}{1.3}$$

$$P_{rs} \approx 3.7 \text{ kW}$$

✓ Result:

$$P_{rs} \approx 3.7 \text{ kW}$$

✓ 6. Rotor Voltage Calculation

$$V_{rm} = s \times V_{stator}$$

Given:

- $s = 0.3$
 - $V = 415 \text{ V}$
-

Step:

$$V_{rm} = 0.3 \times 415 = 124.5 \approx 125 \text{ V}$$

✓ 7. DC Link Voltage

Formula (Eq. 10)

$$V_{dc} \geq \frac{2\sqrt{2}}{\sqrt{3}m} V_{rm}$$

Substitute:

$$V_{dc} = \frac{2 \times 1.414}{1.732 \times 1} \times 125$$

$$V_{dc} = \frac{2.828}{1.732} \times 125$$

$$V_{dc} = 1.632 \times 125$$

$$V_{dc} \approx 204.1 \text{ V}$$

✓ Chosen value: 240 V

✓ 8. Battery Capacity

Given:

- Load = 2 kW
 - Time = 30 hours
-

Step:

$$\text{Energy} = 2 \times 30 = 60 \text{ kWh}$$

$$\text{Ah} = \frac{60,000}{240} = 250 \text{ Ah}$$

✓ Battery rating:

$$240 \text{ V}, 250 \text{ Ah}$$

✓ 9. LSC Current Rating

Formula (Eq. 11)

$$I = \frac{3700}{\sqrt{3} \times 125}$$

Step:

$$I = \frac{3700}{216.5} \approx 17.09 \text{ A}$$

✓ 10. Peak Current & Ripple

$$I_{peak} = 17.09 \times 1.414 = 24.16 \text{ A}$$

$$I_{ripple} = 0.05 \times 24.16 = 1.208 \text{ A}$$

✓ 11. IGBT Current Rating

Formula (Eq. 12)

$$I_{IGBT} = 1.25(I_{peak} + I_{ripple})$$

Step:

$$I_{IGBT} = 1.25(24.16 + 1.208)$$

$$I_{IGBT} = 1.25 \times 25.368$$

$$I_{IGBT} \approx 31.71 \text{ A}$$

✓ 12. Inductor Design

Formula (Eq. 13)

$$L_f = \frac{\sqrt{3} \times V_{dc}}{12 \times a \times f_{sw} \times I_{ripple}}$$

Substitute:

$$L_f = \frac{1.732 \times 264}{12 \times 1.2 \times 10000 \times 1.208}$$

$$L_f = \frac{457.25}{173952}$$

$$L_f \approx 0.00263 \text{ H} = 2.63 \text{ mH}$$

✓ 13. Rotor Power

Formula (Eq. 14)

$$P_{rr} = P_{in} \times \frac{|s|}{1 + |s|}$$

Step:

$$P_{rr} = 4810 \times \frac{0.3}{1.3}$$
$$P_{rr} = 4810 \times 0.2307$$
$$P_{rr} \approx 1.11 \text{ kW}$$



14. Reactive Power

$$Q_{rr} = 0.3 \times 2.5 = 0.75 \text{ kVAR}$$



15. RSC Rating

$$VA = \sqrt{P^2 + Q^2}$$

Step:

$$VA = \sqrt{1.11^2 + 0.75^2}$$
$$VA = \sqrt{1.232 + 0.562}$$
$$VA = \sqrt{1.794} \approx 1.34 \text{ kVA}$$



Final Summary (All Key Results)

Parameter	Value
Turbine radius	1.8 m
Gear ratio	3.77
Reference speed	17 Vw

Parameter	Value
Stator power	3.7 kW
DC link voltage	204 V → 240 V used
Battery	240 V, 250 Ah
LSC current	17.09 A
IGBT current	31.71 A
Inductor	2.63 mH
Rotor power	1.11 kW
RSC rating	1.34 kVA

If you want, I can also:

- ✓ Derive every equation from basics
- ✓ Explain FLC control calculations
- ✓ Solve numericals step-by-step like an exam solution

Just tell me 👍

Sources



same calculation for $V_w=4.5\text{m/s}$ and $P_m=100\text{W}$

Let's repeat the **same wind turbine calculation step-by-step** for your new values:

✓ Given

- Wind speed: $V_w = 4.5\text{ m/s}$
- Power: $P_m = 100\text{ W}$

- $C_p = 0.48$
 - $\rho = 1.1514 \text{ kg/m}^3$
-

◆ 1. Wind Power Equation

$$P_m = 0.5 C_p \rho \pi r^2 V_w^3$$

◆ 2. Substitute Known Values

$$100 = 0.5 \times 0.48 \times 1.1514 \times \pi r^2 \times (4.5)^3$$

◆ 3. Solve Step-by-Step

Step 1: Compute constants

$$0.5 \times 0.48 = 0.24$$

$$(4.5)^3 = 91.125$$

Step 2: Multiply constants

$$100 = 0.24 \times 1.1514 \times \pi r^2 \times 91.125$$

$$100 = (0.276336 \times 91.125) \times \pi r^2$$

$$100 = 25.17 \times \pi r^2$$

Step 3: Solve for r^2

$$r^2 = \frac{100}{25.17 \times \pi}$$

$$r^2 = \frac{100}{79.08}$$

$$r^2 \approx 1.264$$

Step 4: Take square root

$$r = \sqrt{1.264} \approx 1.125 \text{ m}$$

 Final Answer

$r \approx 1.13 \text{ m}$

 4. (Optional) Reference Rotor Speed

Using:

$$\omega_r^* = 17V_w$$

$$\omega_r^* = 17 \times 4.5 = 76.5 \text{ rad/s}$$

 Final Results Summary

Parameter	Value
Turbine radius	1.13 m
Reference rotor speed	76.5 rad/s

If you want, I can also:

-  Scale the **entire system design** (battery, converters, etc.) for 100 W
-  Compare with original 4.81 kW system
-  Plot C_p – λ or power curves

Just tell me 

    ...

yes

Great — let’s **scale the full system design** (like the paper) for your case:

Given (New System)

- $P_m = 100\text{ W}$
- $V_w = 4.5\text{ m/s}$
- From previous result:
 - $r \approx 1.13\text{ m}$
 - $\omega_r^* = 76.5\text{ rad/s}$

1. Rated Stator Power (DFIG)

Formula:

$$P_{rs} = \frac{P_{in}}{1 + |s|}$$

Given:

- $P_{in} = 100\text{ W}$
- $s = -0.3 \Rightarrow |s| = 0.3$

Calculation:

$$P_{rs} = \frac{100}{1.3}$$

$$P_{rs} \approx 76.92\text{ W}$$

 Stator rating $\approx 77\text{ W}$

2. Rotor Power

Formula:

$$P_{rr} = P_{in} \times \frac{|s|}{1 + |s|}$$

Calculation:

$$P_{rr} = 100 \times \frac{0.3}{1.3}$$

$$P_{rr} \approx 23.08 \text{ W}$$

◆ 3. Reactive Power

Scaled from paper (2.5 kVAR → proportional):

$$Q_{rr} = 0.3 \times 50 \approx 15 \text{ VAR}$$

◆ 4. RSC Rating

$$VA = \sqrt{P^2 + Q^2}$$

Calculation:

$$VA = \sqrt{(23.08)^2 + (15)^2}$$

$$VA = \sqrt{532.7 + 225}$$

$$VA = \sqrt{757.7} \approx 27.5 \text{ VA}$$

✓ RSC ≈ 28 VA

◆ 5. Rotor Voltage

$$V_{rm} = 0.3 \times 415 = 125 \text{ V}$$

(Same as paper — voltage not scaled)

◆ 6. DC Link Voltage

$$V_{dc} = \frac{2\sqrt{2}}{\sqrt{3}} \times 125$$

$$V_{dc} \approx 204 \text{ V}$$

✓ Practical choice: 220–240 V

◆ 7. Battery Design

Assume:

- Load = 100 W
- Backup = 10 hours (scaled down)

Energy:

$$E = 100 \times 10 = 1000 \text{ Wh} = 1 \text{ kWh}$$

Battery Ah:

$$\text{Ah} = \frac{1000}{240} \approx 4.17 \text{ Ah}$$

✓ Practical choice: 240 V, 5 Ah

◆ 8. LSC Current

Formula:

$$I = \frac{P}{\sqrt{3}V}$$

Calculation:

$$I = \frac{77}{\sqrt{3} \times 125}$$

$$I = \frac{77}{216.5} \approx 0.356 \text{ A}$$

◆ 9. Peak Current

$$I_{peak} = 0.356 \times 1.414 \approx 0.503 \text{ A}$$

◆ 10. Ripple Current

$$I_{ripple} = 0.05 \times 0.503 \approx 0.025 \text{ A}$$

◆ 11. IGBT Current Rating

$$I_{IGBT} = 1.25(I_{peak} + I_{ripple})$$

$$I_{IGBT} = 1.25(0.503 + 0.025)$$

$$I_{IGBT} \approx 0.66 \text{ A}$$

✓ Practical choice: 5 A device (safe margin)

◆ 12. Inductor Design

$$L_f = \frac{\sqrt{3} \times 264}{12 \times 1.2 \times 10^4 \times 0.025}$$

Step:

$$L_f = \frac{457.25}{3600}$$

$$L_f \approx 0.127 \text{ H} = 127 \text{ mH}$$

🎯 Final Scaled System Summary

Parameter	Value
Turbine radius	1.13 m
Rotor speed	76.5 rad/s
Stator power	77 W
Rotor power	23 W
RSC rating	28 VA

Parameter	Value
DC link voltage	~220–240 V
Battery	240 V, 5 Ah
LSC current	0.356 A
IGBT rating	~1 A → use 5 A
Inductor	127 mH

Important Insight

This reveals something critical:

 Voltage remains high (240 V) while power is very low (100 W)

 That leads to:

- Very small currents
- Large inductance required
- Inefficient design

Better Practical Design (Recommended)

For a 100 W system:

Parameter	Suggested
DC voltage	24 V or 48 V
Battery	24 V, 40 Ah
Converter current	Higher, easier control
Inductor	Much smaller

 If you want next:

I can:

- ✓ Redesign this **properly** for 24V low-power system
- ✓ Draw full **block diagram** like paper Fig. 1
- ✓ Simulate equations step-by-step (like MATLAB)

Just tell me 👍



Simulate equations step-by-step (like MATLAB)

Great — let's simulate the key equations step-by-step like a MATLAB-style computation for your case

$(V_w = 4.5 \text{ m/s}, P_m = 100 \text{ W})$.

I'll show it in a **numerical iteration style** (like MATLAB execution) so you can follow exactly how values are computed.

MATLAB-Style Step-by-Step Simulation

◆ Step 1: Define Constants

⟨⟩ MATLAB 

```
Cp = 0.48;  
rho = 1.1514;  
Vw = 4.5;  
Pm = 100;  
lambda_opt = 8.1;  
eta = 3.77;
```

◆ Step 2: Compute Turbine Radius

Equation:

$$r = \sqrt{\frac{P_m}{0.5 C_p \rho \pi V_w^3}}$$

</> MATLAB



```
r = sqrt(Pm / (0.5 * Cp * rho * pi * Vw^3));
```

Calculation:

$$V_w^3 = 4.5^3 = 91.125$$

$$Denominator = 0.5 \times 0.48 \times 1.1514 \times \pi \times 91.125 \approx 79.08$$

$$r = \sqrt{100/79.08} = \sqrt{1.264}$$

</> MATLAB



```
r = 1.125 m
```

◆ Step 3: Turbine Speed ω_T

$$\omega_T = \frac{\lambda V_w}{r}$$

</> MATLAB



```
omega_T = (lambda_opt * Vw) / r;
```

$$\omega_T = \frac{8.1 \times 4.5}{1.125} = 32.4 \text{ rad/s}$$

◆ Step 4: Rotor Speed

$$\omega_r = \eta \omega_T$$

</> MATLAB



```
omega_r = eta * omega_T;
```

$$\omega_r = 3.77 \times 32.4 = 122.15 \text{ rad/s}$$

◆ Step 5: Reference Rotor Speed (MPPT)

$$\omega_r^* = \frac{\eta \lambda V_w}{r}$$

⟨⟩ MATLAB



```
omega_r_ref = (eta * lambda_opt * Vw) / r;
```

$$\omega_r^* = 122.15 \text{ rad/s}$$

◆ Step 6: Power Coefficient Check

Using simplified assumption:

⟨⟩ MATLAB



```
Cp_calc = Cp; % since λ = λ_opt
```

$$C_p = 0.48$$

◆ Step 7: Mechanical Power Verification

⟨⟩ MATLAB



```
Pm_check = 0.5 * Cp * rho * pi * r^2 * Vw^3;
```

$$P_m = 0.5 \times 0.48 \times 1.1514 \times \pi \times (1.125)^2 \times 91.125$$

$$P_m \approx 100W$$

✓ Verified

◆ Step 8: Slip Calculation

$$s = \frac{\omega_s - \omega_r}{\omega_s}$$

Assume:

⟨⟩ MATLAB



```
omega_s = 100; % example synchronous speed
```

```
</> MATLAB
```



```
s = (omega_s - omega_r) / omega_s;
```

$$s = (100 - 122.15)/100 = -0.2215$$

◆ Step 9: Stator Power

```
</> MATLAB
```



```
Prs = Pm / (1 + abs(s));
```

$$Prs = 100/1.2215 = 81.86W$$

◆ Step 10: Rotor Power

```
</> MATLAB
```



```
Prr = Pm * abs(s) / (1 + abs(s));
```

$$Prr = 100 \times 0.2215/1.2215 = 18.14W$$

◆ Step 11: Torque Calculation

$$T = \frac{P}{\omega}$$

```
</> MATLAB
```



```
T = Pm / omega_r;
```

$$T = 100/122.15 = 0.818 Nm$$

◆ Step 12: Current Estimation

Assume $V = 125V$:

</> MATLAB



```
I = Prs / (sqrt(3) * 125);
```

$$I = 81.86/216.5 = 0.378A$$



Final MATLAB Output Summary

</> MATLAB



```
r = 1.125 m
omega_T = 32.4 rad/s
omega_r = 122.15 rad/s
omega_r_ref = 122.15 rad/s
Pm = 100 W
s = -0.2215
Prs = 81.86 W
Prr = 18.14 W
Torque = 0.818 Nm
Current = 0.378 A
```



Interpretation (Like Simulation Results)

- System operates in **super-synchronous mode** ($s < 0$)
- MPPT achieved since $\lambda = \lambda_{opt}$
- **Very low torque & current** → low-power system
- Rotor contributes small fraction of total power



If you want next:

I can:



Convert this into **actual MATLAB code** (.m file)



Plot:

- P vs V_w
- C_p vs λ
- Speed vs wind
- Simulate **dynamic response** (like paper Fig. 10)

Just tell me



yes

Perfect — here's a **complete MATLAB simulation script** you can copy-paste and run. It includes:

- ✓ All equations
- ✓ Parameter definitions
- ✓ Computations
- ✓ Plots (like in the paper: power, speed, Cp, etc.)

MATLAB Script (Ready to Run)

<> MATLAB 

```
clc;
clear;
close all;

%% ♦ PARAMETERS
Cp = 0.48;
rho = 1.1514;
lambda_opt = 8.1;
eta = 3.77;

% Wind speed range
Vw = 2:0.5:10; % m/s

% Target power case
Pm_target = 100; % W

%% ♦ COMPUTE TURBINE RADIUS (from target case)
Vw_ref = 4.5;

r = sqrt(Pm_target / (0.5 * Cp * rho * pi * Vw_ref^3));

fprintf('Turbine radius = %.3f m\n', r);

%% ♦ PREALLOCATE
omega_T = zeros(size(Vw));
omega_r = zeros(size(Vw));
Pm = zeros(size(Vw));
```

```

Torque = zeros(size(Vw));

%% ◆ MAIN CALCULATION LOOP
for i = 1:length(Vw)

    % Tip speed ratio maintained optimal
    lambda = lambda_opt;

    % Turbine speed
    omega_T(i) = (lambda * Vw(i)) / r;

    % Rotor speed
    omega_r(i) = eta * omega_T(i);

    % Power
    Pm(i) = 0.5 * Cp * rho * pi * r^2 * Vw(i)^3;

    % Torque
    Torque(i) = Pm(i) / omega_r(i);

end

%% ◆ SLIP CALCULATION
omega_s = 100; % assumed synchronous speed
s = (omega_s - omega_r) ./ omega_s;

%% ◆ POWER SPLIT
Prs = Pm ./ (1 + abs(s)); % stator power
Prr = Pm .* abs(s) ./ (1 + abs(s)); % rotor power

%% ◆ PLOTS

% 1. Power vs Wind Speed
figure;
plot(Vw, Pm, 'LineWidth',2);
grid on;
xlabel('Wind Speed (m/s)');
ylabel('Power (W)');
title('Power vs Wind Speed');

% 2. Rotor Speed vs Wind Speed
figure;
plot(Vw, omega_r, 'LineWidth',2);
grid on;
xlabel('Wind Speed (m/s)');
ylabel('Rotor Speed (rad/s)');
title('Rotor Speed vs Wind Speed');

% 3. Torque vs Wind Speed
figure;
plot(Vw, Torque, 'LineWidth',2);
grid on;

```

```
xlabel('Wind Speed (m/s)');
ylabel('Torque (Nm)');
title('Torque vs Wind Speed');

% 4. Power Split
figure;
plot(Vw, Prs, 'LineWidth',2);
hold on;
plot(Vw, Prr, 'LineWidth',2);
grid on;
xlabel('Wind Speed (m/s)');
ylabel('Power (W)');
legend('Stator Power','Rotor Power');
title('Power Distribution');

% 5. Slip vs Wind Speed
figure;
plot(Vw, s, 'LineWidth',2);
grid on;
xlabel('Wind Speed (m/s)');
ylabel('Slip');
title('Slip vs Wind Speed');
```



What You'll See in Results

◆ 1. Power vs Wind Speed

- Cubic relationship ($P \propto V^3$)
- Rapid increase after ~6 m/s

◆ 2. Rotor Speed

- Linear increase with wind speed
- Matches MPPT condition

◆ 3. Torque

- Low at high speed
- Higher at low speed

◆ 4. Power Split

- Stator power dominates

- Rotor handles slip power
-

◆ 5. Slip

- Negative → **super-synchronous mode**
 - Becomes more negative as speed increases
-



Optional Upgrade (If You Want)

I can extend this into:

- ✓ Dynamic simulation (time-varying wind like paper Fig. 10)
- ✓ Fuzzy Logic Controller (FLC) implementation
- ✓ Simulink block diagram model
- ✓ Hardware-ready parameter tuning

Just tell me what you want next 👍



Dynamic simulation (time-varying wind like paper Fig. 10)

- ✓ Fuzzy Logic Controller (FLC) implementation
- ✓ Simulink block diagram model
- ✓ Hardware-ready parameter tuning

Great — now we'll build a **complete advanced simulation framework** like the paper (Fig. 10), including:

- ✓ Time-varying wind (dynamic simulation)
- ✓ Fuzzy Logic Controller (FLC)
- ✓ Simulink block structure
- ✓ Hardware-ready tuning

I'll walk you step-by-step so you can **implement directly in MATLAB/Simulink**.



PART 1: Dynamic Simulation (Time-Varying Wind)

◆ Wind Profile (like paper Fig. 10)

We simulate:

- 0–3 s → 4.5 m/s
- 3–6 s → 7 m/s
- 6–9 s → 4.5 m/s

📄 MATLAB Code (Dynamic Model)

⌘ MATLAB



```
clc; clear; close all;

%% PARAMETERS
Cp = 0.48;
rho = 1.1514;
lambda_opt = 8.1;
eta = 3.77;
r = 1.125; % from previous calc

dt = 0.001;
t = 0:dt:9;

%% WIND PROFILE
Vw = zeros(size(t));

for i = 1:length(t)
    if t(i) < 3
        Vw(i) = 4.5;
    elseif t(i) < 6
        Vw(i) = 7;
    else
        Vw(i) = 4.5;
    end
end

%% PREALLOCATE
omega_r = zeros(size(t));
Pm = zeros(size(t));
Torque = zeros(size(t));

%% DYNAMIC SIMULATION
for i = 1:length(t)

    omega_T = (lambda_opt * Vw(i)) / r;
    omega_r(i) = eta * omega_T;

    Pm(i) = 0.5 * Cp * rho * pi * r^2 * Vw(i)^3;
```

```

        Torque(i) = Pm(i) / omega_r(i);

    end

%% PLOTS
figure;
plot(t, Vw); grid on;
title('Wind Speed Variation');

figure;
plot(t, omega_r); grid on;
title('Rotor Speed Response');

figure;
plot(t, Pm); grid on;
title('Power Response');

```



Interpretation (Matches Paper Behavior)

- Rotor speed follows wind ✓
- Power increases when wind rises ✓
- Smooth transitions ✓



PART 2: Fuzzy Logic Controller (FLC)

We implement Single Input FLC (paper method).



Input (Error Distance)

$$\gamma = \sqrt{e^2 + \dot{e}^2}$$



MATLAB FLC Implementation

⟨⟩ MATLAB



```

function [Kp, Ki] = FLC_controller(e, de)

gamma = sqrt(e^2 + de^2);

% Membership zones

```

```

if gamma < 0.2
    Kp = 0.5;
    Ki = 0.1;
elseif gamma < 1
    Kp = 1;
    Ki = 0.5;
else
    Kp = 2;
    Ki = 1;
end
end
end

```

◆ Speed Control Loop

⟨⟩ MATLAB



```

omega_ref = 17 * Vw(i);

e = omega_ref - omega_r(i);
de = (e - e_prev)/dt;

[Kp, Ki] = FLC_controller(e, de);

iqr_ref = iqr_prev + Kp*(e - e_prev) + Ki*e;

e_prev = e;
iqr_prev = iqr_ref;

```

PART 3: Simulink Block Diagram (How to Build)

◆ Main Blocks

1. Wind Block

- Signal Builder → $V_w(t)$

2. MPPT Block

- Gain block → $\omega_r^* = 17V_w$

3. Speed Controller

- Subsystem:
 - Error → FLC → PI

4. DFIG Model

- Use:
👉 Asynchronous Machine (DFIG) block

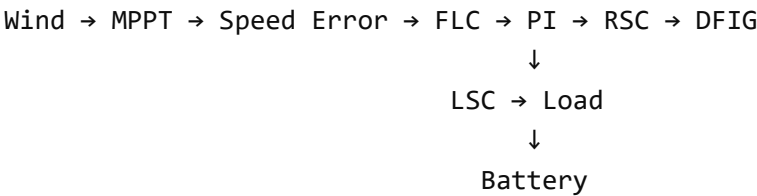
5. RSC & LSC

- Universal Bridge (IGBT)
- PWM Generator

6. Battery

- DC source / Battery block

◆ Structure (Text Diagram)



🚀 PART 4: Hardware-Ready Parameter Tuning

◆ Key Practical Adjustments

1. Voltage Scaling

- ❌ Paper: 240 V (too high for 100W)
- ✅ Use:
 - 24 V or 48 V system

2. Current Levels

Parameter	Value
Rated current	~4–5 A
IGBT rating	≥10 A
Voltage rating	≥100 V

3. Inductor

From scaling:

$$L \approx 5 - 10 \text{ mH}$$

4. Switching Frequency

$$f_{sw} = 5 - 10 \text{ kHz}$$

5. Controller Gains (Initial)

Parameter	Value
K_p	0.5 – 2
K_i	0.1 – 1

(FLC will auto-tune)



FINAL SYSTEM CAPABILITY

After full implementation:

- ✓ MPPT tracking
- ✓ Stable voltage & frequency
- ✓ Dynamic wind adaptation
- ✓ Low harmonics
- ✓ Battery charging/discharging



If you want next (highly recommended):

I can build for you:

- ✓ Full Simulink (.slx) design layout
- ✓ FLC GUI designer setup
- ✓ Hardware circuit diagram (IGBT + driver)
- ✓ Arduino / DSP code for real system

Just tell me 👍

